Solution Proposal: Microservices Migration

Technical Architecture & Modernization Blueprint

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1.0 Problem Statement & Architecture Bottlenecks

The legacy monolithic e-commerce application experiences thread pool exhaustion during flash sales, cascading database lockups, and tight coupling preventing independent team deployments.
Proposed solution: Event-driven microservices architecture leveraging Apache Kafka and Kong Gateway.

2.0 Proposed Target State Architecture

Edge Tier: Cloudflare CDN and Kong API Gateway handling rate limiting, WAF, and SSL offloading.
Application Core: Independent containerized services (Order, Inventory, Customer, Notifications).
Data Tier: Polyglot persistence (PostgreSQL for transactions, Redis for locks, Elastic for search).
Event Streaming: Kafka distributed log for decoupled asynchronous event propagation.

3.0 Phased Implementation Roadmap

Phase 1 (Weeks 1-4): Implement API Gateway and Strangler Fig proxy layer around the monolith.
Phase 2 (Weeks 5-8): Decouple Order & Inventory into microservices with Kafka CDC pipeline.
Phase 3 (Weeks 9-12): Implement OAuth 2.0 Identity Provider and decommission legacy monolith.

Solution Proposal: Microservices Migration - Detailed Specifications

4.0 Cost-Benefit & ROI Analysis

Infrastructure cost optimization: 28% reduction in over-provisioned idle compute servers.
Operational efficiency: Mean-time-to-resolution (MTTR) reduced from 4 hours to under 15 minutes.
Deployment frequency: Increased from bi-weekly release windows to on-demand continuous deployment.